

# M13 Batch 8 — Full Set

## M13.8 Instruments (ATA 31) · PDF pages 651–767

Artificial horizons, attitude director indicators, turn and slip indicators, directional gyros, magnetism and compass systems.

312 questions · Level 3: 91 · Level 2: 148 · Level 1: 73

● marks a question built on a fact your instructor highlighted or underlined in the manual (257 of 312). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

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## Artificial horizons

PDF pages 651–663 · 38 questions

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### LEVEL 1 — RECALL

1. Which two tasks are carried out by the gimballed gyros of an aircraft ●
  - a. attitude measurement and directional measurement
  - b. airspeed measurement and directional measurement
  - c. attitude measurement and vibration measurement
2. What name is given to a gyro whose rotational axis stands upright ●
  - a. rate gyro
  - b. vertical gyro
  - c. directional gyro
3. An attitude gyro is running normally in the cruise. Changes in the angle of which two aircraft axes does it measure? ●
  - a. the longitudinal axis and the vertical axis
  - b. the longitudinal axis and the lateral axis
  - c. the lateral axis and the vertical axis
4. What is the result of the measurement made by the attitude gyro ●
  - a. the angle of flight direction
  - b. the angle of roll and pitch
  - c. the angle of yaw and drift
5. Which gyro gives the reference for the attitude indication of an aircraft ●
  - a. a directional gyro
  - b. a rate gyro
  - c. a vertical gyro
6. Which other name is given to the attitude indication of an aircraft ●
  - a. the artificial horizon
  - b. the attitude director
  - c. the horizon drum
7. A standby horizon is being powered up for a check. From which aircraft supply does its gyro take its drive energy? ●
  - a. the main AC generator
  - b. the battery system
  - c. the cockpit suction pump

8. In what way does the erection system of the standby horizon do its work ●
  - a. pneumatically
  - b. mechanically
  - c. electrically
9. To what maximum rate can an erection system influence the gyro axis ●
  - a. ten degrees per minute
  - b. three degrees per minute
  - c. thirty degrees per minute
10. A technician pulls the caging knob of a standby horizon. What happens inside the instrument? ●
  - a. the gimbals are caged and stabilized in the normal position
  - b. the gyro motor is switched off and allowed to run down
  - c. the erection balls are released onto their circular path
11. What kind of motor is the gyro of a ball driven standby horizon ●
  - a. a three-phase asynchronous internal rotor motor
  - b. a single-phase synchronic external rotor motor
  - c. a three-phase asynchronous external rotor motor
12. The gyro motor of a ball driven standby horizon loses its electrical supply. What shows this on the instrument? ●
  - a. a flag
  - b. a level switch
  - c. an ammeter

## LEVEL 2 — UNDERSTAND AND EXPLAIN

13. A unit on the bench is labelled as a directional gyro. What does that tell a technician about the way its spin axis lies? ●
  - a. it is vertical
  - b. it is horizontal
  - c. it is at 45 degrees
14. A gyro on the bench has its rotational axis standing upright. Which measuring task is it built for? ●
  - a. turn rate measurement
  - b. directional measurement
  - c. attitude measurement
15. Where must the rotational axis of the attitude gyro always point ●
  - a. towards the magnetic north pole
  - b. along the flight path
  - c. towards the centre of the earth
16. Changes in the angle of which axis are determined when direction is measured ●
  - a. the vertical axis
  - b. the lateral axis
  - c. the longitudinal axis
17. A directional gyro is running correctly in level flight. How does its rotational axis lie? ●
  - a. parallel to the horizon of the earth
  - b. along the lateral axis of the aircraft
  - c. straight down at the centre of the earth

18. In a small aircraft, which instrument has the vertical gyro built into it
  - a. the attitude indicator
  - b. the standby horizon
  - c. the suction indicator
19. Why must the artificial horizon stay available for as long as possible ●
  - a. it is the only indication that shows both roll and pitch together
  - b. it is one of the four most important indications in an aircraft
  - c. it is the only indication that works without any electrical power
20. A standby horizon gyro has run down past the point where it still turns stable. What has happened to its indication? ●
  - a. it is frozen at the last attitude
  - b. it is no longer usable
  - c. it is corrected by the caging knob
21. Why does a standby horizon need a caging mechanism as well as its erection system ●
  - a. erection on its own cannot work while the gyro runs at full speed
  - b. erection on its own cannot hold the gimbals in the normal position
  - c. erection on its own is far too slow to set the gyro up at once
22. After which event is an immediate erection of the gyro wanted ●
  - a. a heavy landing
  - b. a long descent
  - c. power activation
23. At which gyro speeds is gimbal caging allowed to be used ●
  - a. at any speed between standstill and full speed
  - b. at full speed or at a complete standstill
  - c. only while the gyro is running down from full speed
24. Apart from the horizon drum, what else is turned by the gyro motor of a ball driven standby horizon ●
  - a. the erection and erection lock mechanism
  - b. the caging knob and the aeroplane symbol
  - c. the pendulum stop and the frame stops
25. What is prevented by the cut-out mechanism of a ball driven standby horizon ●
  - a. the gyroscope tipping towards apparent vertical
  - b. the gyroscope precessing towards earth vertical
  - c. the gyroscope running down after a power failure
26. An aircraft is loaded heavily so it sits nose high on the ground. How can neutral pitch attitude still be set on the standby horizon? ●
  - a. by turning the caging knob to move the aeroplane symbol
  - b. by pulling the caging knob to cage the gimbals
  - c. by resetting the barometric knob of the instrument

**LEVEL 3 — APPLY AND REASON**

27. Two gyros stand on a bench, one with its spin axis upright and one with its spin axis lying flat. Which job does each one do? ●
- the upright one gives roll and pitch, the flat one gives the course
  - the upright one gives the course, the flat one gives roll and pitch
  - the upright one gives pitch only, the flat one gives roll only
28. Every electrical supply in the aircraft has been lost. Why does the standby horizon still give a usable indication for a while? ●
- its erection system needs no power and the gyro runs down slowly
  - its erection system is fed by the standby inverter during the run down
  - its gyro is driven by cockpit air which keeps flowing during the run down
29. A technician cages the gimbals while the gyro is still coasting down at half speed. What is the likely result? ●
- the gyro may be damaged, stability being limited
  - the gyro will stop at once and need a restart
  - the gyro will erect at twice its normal rate
30. A standby horizon has just been switched off and the technician wants to cage its gimbals. What should be done first? ●
- pull the caging knob twice in succession
  - wait until the gyro is stopped completely
  - cage at once before the speed starts to fall
31. The gyro axis points at the centre of the earth during unaccelerated flight. What are the two erection balls doing? ●
- they are balanced out, so no momentum acts on the gyro axis
  - they are driven apart, so a momentum acts on the gyro axis
  - they are held at their stops by the gyro cut-out rocker
32. A horizontal acceleration greater than 1.7 m/s<sup>2</sup> acts on a ball driven standby horizon. What happens inside it? ●
- a flag drops to show the indication is no longer usable
  - a torque motor drives the balls apart to correct the axis
  - a rocker tips and prevents an unbalance of the balls
33. At start-up the gyro axis of a ball driven horizon is found more than 10 degrees out of earth vertical. What must be done? ●
- switch the supply off and let the gyro run down
  - pull the caging knob to erect the gyro rapidly
  - wait for the erection balls to bring the axis back
34. Power to a ball driven standby horizon is lost in flight. For how long can the instrument still be used? ●
- up to 15 minutes
  - up to 45 minutes
  - up to 5 minutes

35. A ball driven standby horizon is switched on with its axis only slightly off earth vertical. Why does it still erect on its own? ●
- the erection balls run faster below full gyro speed
  - the caging knob is pulled by the power up circuit
  - the cut-out only blocks erection beyond 10 degrees
36. A ball driven standby horizon is held in a long steady horizontal acceleration. Why does its erection stop working? ●
- a flag drops, so the erection supply is cut off
  - the caging knob latches the gimbals in position
  - a rocker tips, so the balls cannot go out of balance
37. A ball driven standby horizon is fed from the battery through a static inverter. What kind of supply does its gyro motor need? ●
- single-phase DC
  - half-wave DC
  - three-phase AC
38. A standby horizon has lost all power in flight. What sets the limit on how long the crew may keep using it? ●
- the rotational energy still stored in the gyro
  - the air still being drawn through the instrument
  - the charge still left in the aircraft battery

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### Answers — Artificial horizons

- a** The two important functions named for gimballed gyros are attitude measurement and directional measurement. (L1, PDF p. 651)
- b** A gyro with a vertical axis is called a vertical gyro. (L1, PDF p. 651)
- b** The attitude gyro measures the angle changes of the x-axis and the y-axis. (L1, PDF p. 653)
- b** Measuring the x and y axis angle changes gives the angle of roll and pitch. (L1, PDF p. 653)
- c** The vertical gyro is the reference for the attitude indication. (L1, PDF p. 657)
- a** The attitude indication is also called the artificial horizon. (L1, PDF p. 657)
- b** The standby horizon gyro is electrically driven from the aircraft battery system. (L1, PDF p. 659)
- b** The erection system of the standby horizon works mechanically. (L1, PDF p. 659)
- b** An erection system can only influence the gyro axis to a maximum of three degrees per minute. (L1, PDF p. 661)
- a** Pulling the caging knob cages the gimbals and stabilizes them in the normal position. (L1, PDF p. 661)
- c** The gyro is a three-phase asynchronous external rotor motor. (L1, PDF p. 663)
- a** A flag monitors the electrical supply to the motor. (L1, PDF p. 663)
- b** Gyros with a horizontal axis are the ones called directional gyros. (L2, PDF p. 651)
- c** The vertical axis gyro is the attitude gyro, so it does attitude measurement. (L2, PDF p. 651)
- c** The attitude gyro axis must always point at the centre of the earth, that is, be vertical. (L2, PDF p. 653)
- a** Direction measurement comes from the angle changes of the vertical axis. (L2, PDF p. 653)
- a** The directional gyro axis must always be parallel to the earth's horizon, that is horizontal. (L2, PDF p. 653)
- a** In small aircraft the vertical gyro sits inside the attitude indicator. (L2, PDF p. 657)
- b** It is named as one of the four most important indications, so it must keep working as long as it can. (L2, PDF p. 659)
- b** The indication stays usable only as long as the gyro runs stable during its spin down time. (L2, PDF p. 659)
- c** Erection moves the axis only three degrees a minute, so a caging mechanism is needed to erect the gyro immediately. (L2, PDF p. 661)

- 22. c Immediate erection is wanted for example after power activation. (L2, PDF p. 661)
- 23. b Caging may be used only with the gyro at full speed or stopped completely. (L2, PDF p. 661)
- 24. a The same motor also drives the erection and the erection lock mechanism. (L2, PDF p. 663)
- 25. a The cut-out mechanism stops the gyroscope from tipping towards apparent vertical. (L2, PDF p. 663)
- 26. a Turning the caging knob raises or lowers the symbol, so neutral pitch can be set whatever the weight. (L2, PDF p. 663)
- 27. a The vertical axis gyro is the attitude gyro giving roll and pitch; the horizontal axis gyro is the directional gyro giving the course. (L3, PDF p. 653)
- 28. a The mechanical erection system keeps working with no power while the gyro coasts down, so the indication stays usable. (L3, PDF p. 659)
- 29. a Any intermediate speed may damage the gyro because of the limited stability. (L3, PDF p. 661)
- 30. b Caging is allowed only at full speed or at a complete standstill, so after switch off the gyro must be left to stop. (L3, PDF p. 661)
- 31. a With the axis in earth vertical the balls are balanced and no momentum acts on the gyro axis. (L3, PDF p. 663)
- 32. c Above 1.7 m/s<sup>2</sup> a rocker tips and keeps the balls from going out of balance. (L3, PDF p. 663)
- 33. b Beyond 10 degrees the cut-out mechanism blocks independent erection, so the caging knob must be pulled. (L3, PDF p. 663)
- 34. a The gyrating mass lets the standby horizon be used for up to 15 minutes after a power failure. (L3, PDF p. 663)
- 35. c The cut-out mechanism prevents independent erection only when the axis is more than 10 degrees out of earth vertical. (L3, PDF p. 663)
- 36. c Above 1.7 m/s<sup>2</sup> the rocker of the pendulum accelerometer tips and prevents the ball unbalance that would erect the gyro. (L3, PDF p. 663)
- 37. c The gyro is a three-phase motor, which is why the static inverter is fitted between it and the battery. (L3, PDF p. 663)
- 38. a The gyrating mass carries the gyro on for up to 15 minutes, and that stored energy is what sets the limit. (L3, PDF p. 663)

## Vertical gyros and erection systems

PDF pages 665–679 · 66 questions

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### LEVEL 1 — RECALL

39. How far can the mass on one end of the balance beam swing out when the gyro is unbalanced ●
- a. up to 10 degrees
  - b. up to 85 degrees
  - c. up to 30 degrees
40. In a small aircraft, what generates the airflow that drives the gyro ●
- a. a vacuum or suction pump
  - b. a static inverter
  - c. an engine bleed tapping
41. For what do the letters ISIS stand ●
- a. Internal Standby Indicating System
  - b. Integrated Standby Instrument System
  - c. Integrated Second Instrument Set
42. Which three indications are supplied by a recent integrated standby instrument system ●
- a. attitude, heading and mach number
  - b. attitude, altitude and airspeed
  - c. altitude, airspeed and vertical speed
43. How many degrees of freedom does a vertical gyro have ●
- a. one
  - b. two
  - c. three
44. How are vertical gyros driven ●
- a. by cockpit airflow
  - b. electrically
  - c. by engine bleed air
45. Which switches do electrical gyros carry to deal with aircraft accelerations ●
- a. gimbal caging switches
  - b. erection cutout switches
  - c. conductive level switches
46. Which aircraft axis does the roll axis of the vertical gyro correspond to ●
- a. the lateral axis
  - b. the longitudinal axis
  - c. the vertical axis
47. Which aircraft axis is represented by the pitch axis of the vertical gyro ●
- a. the vertical axis
  - b. the lateral axis
  - c. the longitudinal axis

48. To what rate is the normal speed of erection of the vertical gyro limited ●
- 2 degrees per minute
  - 20 degrees per minute
  - 12 degrees per minute

## LEVEL 2 — UNDERSTAND AND EXPLAIN

49. How does an improved standby horizon obtain the alternating current it needs to run
- from a phase converter mounted in the aircraft rack
  - from a static inverter installed inside the device
  - from the standby busbar of the aircraft supply
50. Which mechanism is absent from a pendulum erection standby horizon ●
- the gimbal caging mechanism
  - the ball circulating mechanism
  - the mechanical cut-out mechanism
51. What takes the place of the ball circulating mechanism in a pendulum erection horizon ●
- a balance beam turning at low speed
  - a pendulum accelerometer and rocker
  - a pair of conductive level switches
52. The gyro of a standby horizon with a balance beam is no longer in its horizontal position and the beam mass swings out. What is that displacement used for? ●
- as the erection momentum for the gyroscope
  - as the signal that drops the warning flag
  - as the braking force on the gyro capsule
53. Besides a large gyro unbalance, what else makes the cut-out mechanism hold the rotational mass in place ●
- equivalent horizontal accelerations
  - a drop of the supply voltage
  - a rise of the gyro rotational speed
54. Why does a gyro need a high rotational speed
- to lower its power consumption
  - to shorten its erection time
  - to have enough stability
55. The suction pump of a small aircraft is running. Which air is being drawn through the indicators? ●
- the bleed air, through a cooling unit
  - the cockpit air, through a filter
  - the ram air, through an intake scoop
56. What shape is given to the gyro body of an air driven instrument ●
- it is formed like a small turbine
  - it is formed like a flat disc
  - it is formed like a hollow sphere
57. Which two advantages does an electric motor drive give to the gyros of commercial aircraft ●
- more reliability and a stable speed in all flight situations
  - a simpler filter fit and a lower running temperature
  - lower weight and a shorter erection time after switch on

58. In what kind of construction are the latest standby horizons built
- a mechanically caged standby gyro with a horizon drum
  - an air driven standby gyro with pendulum erection
  - a solid state electronic standby instrument system
59. Besides attitude, altitude and airspeed, which two further values can a recent integrated standby system display ●
- magnetic heading and mach number
  - drift angle and ground speed
  - outside air temperature and wind
60. An aircraft is fitted with an integrated standby instrument system. What has become of the separate standby instruments? ●
- they are kept as a second back up
  - they are moved to the overhead panel
  - they are no longer needed
61. In which case is a vertical gyro used for an artificial horizon ●
- when the gyro must be driven by cockpit airflow
  - when the gyro has no erection system of its own
  - when the gyro is not linked mechanically to the dial
62. A vertical gyro is mounted in the avionic compartment as an attitude sensor. What else must the installation then have? ●
- a suction pump in the cockpit
  - an indicator in the cockpit
  - a caging knob in the compartment
63. What holds the axis of a vertical gyro in the vertical position ●
- a caging mechanism
  - an erection system
  - a pendulum stop
64. The aircraft rolls while a remote vertical gyro is working. By what is the angle of the attitude sensed? ●
- by torque motors
  - by level switches
  - by synchros
65. What does an air driven gyro use to generate its erection force ●
- a balance beam turning slowly
  - pendulums moved by the airflow
  - level switches holding a fluid
66. What activates the torque motor of an electrically driven gyro when its axis topples ●
- a pendulum moved by the cockpit airflow
  - a level switch holding a conductive fluid
  - a rocker tipped by horizontal acceleration
67. A force has to be applied to move a gyro axis in a chosen direction. At what angle from that direction must the force act? ●
- 180 degrees
  - 45 degrees
  - 90 degrees

68. Which error must the normal erection rate compensate, besides precessions from real topple ●
- a real topple of 20 degrees per minute
  - an erection drift of 3 degrees per minute
  - an apparent topple of 15 degrees per hour
69. An electrical gyro is switched on again after a long power interruption. Which erection rate is used? ●
- about 3 degrees per minute
  - about 20 degrees per minute
  - about 2 degrees per minute
70. What can a vertical gyro no longer detect once longitudinal accelerations or centrifugal forces act on it ●
- the lateral axis of the aircraft
  - its own rotational speed
  - the centre of the earth
71. During which manoeuvres must the erection system be switched off ●
- during curves or longitudinal accelerations
  - during climbs and descents at steady speed
  - during taxiing and engine ground running
72. An erection cutout switch has sensed a roll axis input while the aircraft manoeuvres. For what period does it keep the erection circuit interrupted? ●
- for as long as the acceleration lasts
  - for a fixed period of three seconds
  - until the gyro is caged by the technician
73. Why must long acceleration periods such as full turns be avoided ●
- the erection rate rises above its normal working limit
  - the gyro motor runs hot in that attitude of the frame
  - the gyro topple cannot be compensated during that time
74. When the attitude of the aircraft changes, what turns around what ●
- the aircraft turns around the gyroscope
  - the gyroscope precesses together with the aircraft
  - the gyroscope turns around the aircraft
75. How do the gyro axis and the frame axis lie to one another in horizontal flight ●
- at right angles to one another
  - parallel to one another
  - at 45 degrees to one another
76. A vertical gyro is feeding the cockpit indicator. What turns the angular movements at the pivot points into electric signals? ●
- synchro-transmitters
  - erection torque motors
  - conductive level switches
77. What are the voltages between the tapped synchro wires of a locked synchro when the aircraft is in horizontal attitude
- at half of full value
  - zero
  - at a maximum

- 78.** Which two users take the synchro signals of the vertical gyro ●
- the flight director and the weather radar unit
  - the attitude director indicator and the autopilot
  - the weather radar and the flight director unit
- 79.** Which two users are switched to the locked synchros of the vertical gyro ●
- the autopilot and the weather radar
  - the weather radar and the flight director
  - the attitude director indicator and the autopilot
- 80.** With which two means can the alignment of the gyro axis be measured
- a plumb line or a spirit level
  - a synchro or a pendulum
  - a level switch or a torquer

### LEVEL 3 — APPLY AND REASON

- 81.** A pendulum erection horizon is sitting with a gyro unbalance of 12 degrees. What does its cut-out mechanism do? ●
- it holds the rotational mass in its normal position
  - it lets the rotational mass swing out to full travel
  - it breaks the electrical supply to the gyro motor
- 82.** A standby horizon is found with no AC feed at all, only 28 V DC. Which type is it and why does it run?
- a ball driven type, its inverter sits inside the device
  - a pendulum erection type, its gyro motor runs straight off DC
  - a pendulum erection type, its inverter sits inside the device
- 83.** The suction regulator of an air driven gyro system fails and the suction rises well above its normal value. What is the danger to the gyro?
- it will topple to apparent vertical
  - it will run down earlier than usual
  - it will run into an overspeed
- 84.** A commercial aircraft gyro and a small aircraft gyro are compared. Which has its speed held steady in every flight situation, and why? ●
- the commercial one, because its drive is a filtered cockpit airflow
  - the small one, because its drive comes through a suction regulator
  - the commercial one, because its drive is an electric motor
- 85.** A gyro instrument is found with a filter and a suction line behind it. In which class of aircraft is it fitted? ●
- a small aircraft
  - a commercial aircraft
  - an aircraft with no battery
- 86.** A vertical gyro sits in the avionic compartment and its synchros are working normally. What makes use of the signal they send? ●
- the indicator or an indication system
  - the erection cutout switches
  - the supply converter of the gyro motor

87. An artificial horizon is built with its gyro in a remote box and a separate indicator in the cockpit. Which gyro is in the box and how is the attitude passed on? ●
- a directional gyro, passed on by synchros
  - a vertical gyro, passed on by synchros
  - a vertical gyro, passed on by a linkage
88. The roll level switch of an electrically driven gyro detects the axis in an incorrect direction. What follows? ●
- the roll torque motor applies a force and precession corrects it
  - the caging mechanism locks the gimbals in a normal position
  - the roll cutout switch opens and the erection circuit is broken
89. An aircraft makes normal turns and accelerations in the cruise. Why does the erection system build up no significant error? ●
- because the rate it works at is doubled
  - because the rate it works at is set by hand
  - because the rate it works at is so small
90. A vertical gyro has stood without power for a long time and its axis is well off the vertical. Why is a special erection rate needed to get it ready? ●
- the normal erection rate works only on the ground
  - the normal erection rate would be far too slow
  - the normal erection rate would damage the gimbals
91. An air driven gyro and an electrically driven gyro are both off the vertical. Which part supplies the correcting force in each? ●
- a balance beam in the one, erection balls in the other
  - level switches in the one, pendulums in the other
  - pendulums in the one, a torque motor in the other
92. Aircraft in a hold fly racetrack patterns rather than continuous turns. What does the shape of that pattern give the erection system? ●
- time after each half turn to compensate gyro topple
  - a steady centrifugal force to erect the gyro against
  - a continuous cutout of both of the erection circuits
93. An aircraft is held in one long continuous turn instead of a racetrack. What happens to the attitude reference? ●
- the gyro erects faster because of the bank angle
  - the gyro is caged automatically until the turn ends
  - the gyro topples and the error is not corrected
94. An aircraft settles into a steady turn. What happens to the erection of its vertical gyro, and what follows from it? ●
- the torquer is driven harder, so the axis is held upright
  - the fast rate is switched in, so the axis is held upright
  - the circuit is cut, so any topple is left uncorrected
95. The attitude of the aircraft changes while a vertical gyro is working. Which part of the synchro moves with the airframe? ●
- both stator and rotor, around the fixed gyro case
  - the stator, around the rotor held steady by the gyro
  - the rotor, around the stator held steady by the gyro

96. A deviation voltage from a locked synchro changes its phase and grows in amplitude. What do those two changes each tell you? ●
- the phase gives the direction and the amplitude the size
  - the phase gives the size and the amplitude the direction
  - the phase gives the axis and the amplitude the rate
97. A vertical gyro is sitting on the shelf with no power on it. Which stop keeps the gyroscope out of an extreme inclination?
- the pitch stop
  - the pendulum stop
  - the frame stop
98. A vertical gyro has been taken to about 85 degrees of movement around the lateral axis of the aircraft. What holds it there?
- the pendulum stop
  - the frame stop
  - the pitch stop
99. The air bubbles of the level switches have wandered out of the centre of their tubes. What then happens at the torquer?
- the currents stay equal and no torque is produced
  - the currents become unequal and torque is produced
  - the currents are broken and the circuit is deactivated
100. Why do equal currents in the two variable windings of the torquer leave the gyro axis untouched?
- the magnetic fields are unequal, so the two torques cancel out
  - the magnetic fields are equal and opposite, so no torque results
  - the magnetic fields are equal and additive, so no torque results
101. An erection circuit stays active while the aircraft accelerates. Where is the gyro led, and what does the crew see?
- towards true perpendicular, and a flag in place of the display
  - towards apparent vertical, and an equally large error on the display
  - towards the pendulum stop, and a frozen reading on the display
102. The roll cut out switches of a gyro have failed open circuit so they never cut the erection. Where will the gyroscope be moved during a turn?
- into apparent vertical
  - into true perpendicular
  - onto the pendulum stop
103. A gyro responds 90 degrees away from an applied force. How is that allowed for in the way the torquers are connected?
- the pitch torquer puts its torque on the roll axis
  - the pitch torquer puts its torque on the pitch axis
  - the pitch torquer puts its torque on the yaw axis
104. An aircraft is standing still on the ramp with its vertical gyro powered. Why may the erection circuit work freely here?
- only the gyro is turning below its normal speed
  - only the weight is acting as an acceleration
  - only the cut out switches are disconnected here

## Answers — Vertical gyros and erection systems

39. c The mass on one end can swing out up to 30 degrees in the case of gyro unbalance. (L1, PDF p. 665)
40. a The airflow comes from a vacuum or suction pump. (L1, PDF p. 667)
41. b ISIS is the Integrated Standby Instrument System. (L1, PDF p. 669)
42. b Recent systems supply attitude, altitude and airspeed indications. (L1, PDF p. 669)
43. c A vertical gyro is a gyro with three degrees of freedom. (L1, PDF p. 671)
44. b Vertical gyros are driven electrically. (L1, PDF p. 671)
45. b Electrical gyros have erection cutout switches for this. (L1, PDF p. 675)
46. b The roll axis of the gyro corresponds to the aircraft's longitudinal axis. (L1, PDF p. 677)
47. b The pitch axis of the gyro represents the lateral axis of the aircraft. (L1, PDF p. 677)
48. a Normal speed of erection is limited to 2 degrees per minute. (L1, PDF p. 679)
49. b A static inverter built into the device supplies the AC needed for operation. (L2, PDF p. 665)
50. b These devices do not have a ball circulating mechanism. (L2, PDF p. 665)
51. a In its place a balance beam rotates at low speed. (L2, PDF p. 665)
52. a The mass displacement is used as erection momentum for the gyroscope. (L2, PDF p. 665)
53. a Equivalent horizontal accelerations have the same effect as a 10 degree unbalance. (L2, PDF p. 665)
54. c High rotational speed is what gives the gyro enough stability. (L2, PDF p. 667)
55. b The pump sucks cockpit air via a filter through the indicators. (L2, PDF p. 667)
56. a The gyro body is formed like a small turbine so the airflow can drive it. (L2, PDF p. 667)
57. a Electric drive is stated to be more reliable and to give a stable speed in all flight situations. (L2, PDF p. 667)
58. c The latest standby horizons are integrated into a solid state electronic standby instrument system. (L2, PDF p. 669)
59. a Magnetic heading and mach number can be displayed in addition. (L2, PDF p. 669)
60. c With the integrated system the individual original standby instruments are no longer needed. (L2, PDF p. 669)
61. c The vertical gyro is used where the gyro is not mechanically linked to an instrument's indication. (L2, PDF p. 671)
62. b With the gyro in the avionic compartment an indicator is needed in the cockpit. (L2, PDF p. 671)
63. b An erection system holds the gyro axis vertical. (L2, PDF p. 671)
64. c The angle of attitude is sensed by synchros. (L2, PDF p. 671)
65. b The air driven gyro uses pendulums to make an erection force from the airflow. (L2, PDF p. 673)
66. b Electrically driven gyros use level switches with a conductive fluid to activate the torque motor. (L2, PDF p. 673)
67. c Precession means forces must be applied 90 degrees from the result to be achieved. (L2, PDF p. 673)
68. c The rate is sufficient to compensate the apparent topple of 15 degrees per hour plus precessions from real topple. (L2, PDF p. 673)
69. b A faster erection rate of about 20 degrees per minute is activated after a long power interruption. (L2, PDF p. 673)
70. c Under those forces the gyro can no longer detect the centre of the earth. (L2, PDF p. 675)
71. a The erection system must be switched off during curves or longitudinal accelerations. (L2, PDF p. 675)
72. a The switch interrupts the corresponding erection circuit for as long as the accelerations last. (L2, PDF p. 675)
73. c During a long acceleration the erection is cut out, so gyro topple cannot be compensated. (L2, PDF p. 675)
74. a Because the gyro is space stable, any change in attitude is a rotating of the aircraft around the gyroscope. (L2, PDF p. 677)
75. a In horizontal flight the gyro axis and the frame axis are at right angles. (L2, PDF p. 677)
76. a Synchro-transmitters at the pivot points make the angular movements into electric signals. (L2, PDF p. 677)
77. b Those voltages are zero at horizontal attitude of the aircraft. (L2, PDF p. 677)
78. b The ADI and the autopilot use the synchro signals. (L2, PDF p. 677)
79. b WX and FD are switched to the locked synchros. (L2, PDF p. 677)
80. a The alignment can be measured with a plumb line or a spirit level. (L2, PDF p. 679)

81. a At an unbalance of 10 degrees or more the cut-out mechanism retains the rotational mass in its normal position. (L3, PDF p. 665)
82. c Only the improved pendulum erection horizon is fed with 28 V DC alone, because the static inverter that makes its AC is installed inside the device. (L3, PDF p. 665)
83. c The regulator is there to prevent gyro overspeed, so losing its control lets the gyro overspeed. (L3, PDF p. 667)
84. c Commercial aircraft use electrically driven gyros, which give a stable speed in all flight situations. (L3, PDF p. 667)
85. a Air driven gyros with a suction pump and filter belong to small aircraft, while commercial aircraft use electric drive. (L3, PDF p. 667)
86. a The synchro signal goes to the cockpit indicator or to an indication system. (L3, PDF p. 671)
87. b A vertical gyro is used where there is no mechanical link, and the attitude angle it senses goes to the indicator as a synchro signal. (L3, PDF p. 671)
88. a The roll level switch activates the roll torque motor, whose force precesses the gyro back into the correct direction. (L3, PDF p. 673)
89. c The erection rate is so small that normal turns and accelerations cannot build a significant error. (L3, PDF p. 673)
90. b Normal erection is far too slow to bring a badly toppled axis back, so a faster erection rate of about 20 degrees per minute is switched in after a long power interruption. (L3, PDF p. 673)
91. c The air driven gyro erects with pendulums in the airflow, while the electrical gyro erects with a torque motor fed through level switches. (L3, PDF p. 673)
92. a The racetrack gives the erection system enough time after each half turn to compensate any gyro topple. (L3, PDF p. 675)
93. c The cutout switches hold the erection circuit open throughout the turn, so gyro topple builds up uncorrected. (L3, PDF p. 675)
94. c The roll cutout switch interrupts the erection circuit for as long as the turn lasts, so gyro topple cannot be compensated. (L3, PDF p. 675)
95. b The synchro-stator turns with the aircraft around the synchro-rotor, which the gyroscope keeps stabilised. (L3, PDF p. 677)
96. a Phase shift represents the direction of the deviation and amplitude its dimension. (L3, PDF p. 677)
97. b The pendulum stop is the one that prevents an extreme inclination when the gyro is not energised. (L3, PDF p. 677)
98. c The pitch stop limits movement around the lateral axis to about +/- 85 degrees. (L3, PDF p. 677)
99. b Off centre bubbles make the currents in the variable windings unequal, so the torquer produces a torque. (L3, PDF p. 679)
100. b With the same currents the magnetic fields are equal but opposite, and the torquer produces no torque. (L3, PDF p. 679)
101. b With erection active during acceleration the gyro is led to apparent vertical, and the axis deviation shows as an equally large error on the horizon display. (L3, PDF p. 679)
102. a The roll cut out switches are there to prevent the gyroscope being moved into apparent vertical, so without them the turn leads it there. (L3, PDF p. 679)
103. a To use precession, the pitch torquer is made to act on the roll axis and the roll torquer on the pitch axis. (L3, PDF p. 679)
104. b In a stationary condition only the earth's acceleration has an effect, which is the condition the erection circuit needs. (L3, PDF p. 679)

## Attitude director indicators

PDF pages 681–693 · 50 questions

### LEVEL 1 — RECALL

105. Which supply voltage is fed to the vertical gyro ●
  - a. 90 V AC
  - b. 28 V DC
  - c. 115 V AC
106. How long does the fast erection phase last after the vertical gyro is switched on ●
  - a. about 2 - 3 minutes
  - b. about 30 seconds
  - c. a maximum of three minutes
107. Where is the vertical gyro that supplies the attitude signals to the ADI installed ●
  - a. inside the ADI housing itself
  - b. in the electric compartment
  - c. in the cockpit overhead panel
108. How many independent vertical gyros are needed so that each pilot has an attitude indication of his own ●
  - a. one
  - b. two
  - c. three
109. Which other aircraft systems are supplied with attitude information from the vertical gyros ●
  - a. the autoflight system and the weather radar system
  - b. the instrument landing system and the radio altimeter
  - c. the air data system and the instrument warning system
110. Which two kinds of information does the attitude director indicator mainly show ●
  - a. attitude and indications from the instrument landing system
  - b. heading and indications from the flight director system
  - c. attitude and indications from the flight director system
111. To which group of main flight instruments does the attitude director indicator belong ●
  - a. the pitot-static group
  - b. the Basic T
  - c. the engine indicating group
112. What kind of instrument is the slip indicator fitted to the ADI ●
  - a. an electric servo motor fed from the vertical gyro
  - b. a mechanical ball in a liquid filled curved glass tube
  - c. a moving coil movement fed with the yaw rate signal
113. Which rate of turn movement does the turn indication on the ADI represent ●
  - a. 180°/min
  - b. 360°/min
  - c. 90°/min

- 114.** Why is the attitude comparison fitted to the attitude reference system ●
- a. as a means of erecting the gyros faster
  - b. as an additional safety feature
  - c. as a replacement for the third gyro

## LEVEL 2 — UNDERSTAND AND EXPLAIN

- 115.** Which condition must have applied before switch on for the vertical gyro to run a fast erection phase ●
- a. the gyro had been without current for more than 30 seconds
  - b. the erectors had been held at the reduced support rate
  - c. the gyro motor had been running above 75% rpm
- 116.** A vertical gyro has held a motor speed above 75% for more than 30 seconds and its gimbals are erected. What happens next? ●
- a. fast erection begins and the motor runs on at raised voltage
  - b. the erectors are switched off for a maximum of three minutes
  - c. fast erection ends and the motor runs on at reduced voltage
- 117.** Which erector is switched off by a strong acceleration along the aircraft's length, and which by a strong horizontal acceleration? ●
- a. the roll erector, and the pitch erector
  - b. the pitch erector, and the roll erector
  - c. the pitch erector in both of the two cases
- 118.** What is the longest time the pitch erector may stay switched off during strong longitudinal acceleration ●
- a. three minutes
  - b. two minutes
  - c. 30 seconds
- 119.** A flag monitor is checking the gyro supply before it releases the valid voltage. What level must that supply exceed?
- a. 75 V
  - b. 115 V
  - c. 90 V
- 120.** A technician traces the ADI pitch and roll signals back to the vertical gyro gimbals. What will he find converting the angles into electrical signals? ●
- a. servo motors
  - b. moving coils
  - c. synchros
- 121.** An ADI receives a gyro valid signal from its vertical gyro. What has the gyro confirmed by sending it?
- a. that its gimbals are free of friction
  - b. that its synchros are correctly aligned
  - c. that it is running at its normal speed
- 122.** Which signal reaches the ADI only once its vertical gyro is up to normal speed and fast erection is complete ●
- a. the roll synchro signal
  - b. the pitch synchro signal
  - c. the valid signal

123. One of the two normal vertical gyros has failed in flight. How is that pilot's attitude indication restored? ●
- his indicator falls back on the standby horizon signal
  - a switching transfers the third vertical gyro to his indicator
  - his indicator is cross-wired to the other pilot's vertical gyro
124. A gyro has failed and the spare is to be brought in. To which indicators can that third gyro be switched? ●
- to both ADIs at the same time
  - to one fixed ADI only
  - to either of the two pilots' ADIs
125. A student asks why the aeroplane symbol of the ADI follows every movement of the aircraft. What is the answer? ●
- it is carried on the instrument housing itself
  - it is driven by the roll synchro of the vertical gyro
  - it is mounted on the fork that holds the horizon ball
126. How does the blue and black horizon ball of the ADI behave as the aircraft manoeuvres
- it turns with the aircraft movement and is kept horizontal by the gyro
  - it is held still while the pitch and roll scales move around it
  - it turns contrary to the aircraft movement and is kept horizontal by the gyro
127. What does the VG valid direct current arriving at the ADI confirm ●
- that the synchro signals have nulled
  - that the vertical gyro is functioning
  - that the ADI power supply is healthy
128. What makes the ball of the slip indicator take up the apparent vertical ●
- the yaw rate signal fed to a coil behind the ball
  - the deviation signal of the navigation receiver
  - the influence of gravity on the ball
129. Which aircraft movement supplies the signal for the rate of turn part of the ADI ●
- the pitch rate
  - the yaw rate
  - the roll rate
130. An aircraft's rotation changes while the ADI turn indicator is being watched. Which change does that movement answer? ●
- a change in rotational velocity about the lateral axis
  - a change in rotational velocity about the vertical axis
  - a change in the resultant force acting on the aircraft
131. Which kind of movement drives the turn indication on the ADI ●
- a synchro receiver
  - a moving coil
  - an electric servo motor
132. How much deviation from the localizer do the marks beside the runway symbol stand for
- about one degree left or right
  - about three degrees left or right
  - about one third of a degree left or right

- 133.** A technician must identify the drive behind the glideslope pointer of an ADI. What kind of movement will he find? ●
- a synchro receiver
  - an electric servo motor
  - a moving coil
- 134.** What do the flight director bars show the pilot ●
- the direction in which the aircraft is to be flown
  - the deviation of the aircraft from the runway centreline
  - the direction in which the aircraft is actually moving
- 135.** From where does the attitude comparison take the signals it works on
- directly from the vertical gyro gimbals
  - directly from the indication
  - directly from the autoflight system
- 136.** How far apart must the pitch or roll indications be before the alert is raised ●
- any amount apart at all
  - several minutes of arc apart
  - several degrees apart
- 137.** Which system raises the alert when the two attitude indications disagree ●
- the instrument warning system
  - the weather radar system
  - the autoflight system
- 138.** An aircraft has primary flight displays in place of ADIs and the two attitudes disagree. What do the crew see? ●
- a CHECK ATTITUDE message above the attitude indication on one display
  - an attitude flag across the attitude indication on both displays
  - a CHECK ATTITUDE message above the attitude indication on both displays
- 139.** Which sources can serve as the third attitude reference for fault isolation ●
- vertical gyro number 1 or the standby horizon
  - vertical gyro number 3 or the weather radar
  - vertical gyro number 3 or the standby horizon

### LEVEL 3 — APPLY AND REASON

- 140.** A vertical gyro has just reported valid and the voltage on its erectors has dropped. What support rate are the erectors working at now? ●
- 2°/min
  - 28°/min
  - 20°/min
- 141.** Shortly after power up a vertical gyro is reporting invalid while its erectors run at the higher support rate. What does this tell a technician? ●
- the gyro has finished erecting and is in normal erection
  - the gyro motor has failed to reach 75% rpm
  - the gyro is still in its fast erection phase

- 142.** A vertical gyro is at full speed with its output synchros energised, yet the flag monitor still withholds the valid voltage. Which remaining condition is missing?
- the supply has not yet reached its rated alternating voltage
  - the roll erector has not yet been switched off
  - fast erection has not yet been completed
- 143.** An aircraft carries a second autopilot as well as its original one. Which vertical gyro feeds that second system? ●
- the first officer's vertical gyro
  - the third vertical gyro
  - the captain's vertical gyro
- 144.** The aircraft rolls into a bank. How does the horizon ball of the ADI answer that movement?
- the fork swings the ball the opposite way round
  - the fork and the ball are both held still
  - the ball spins about its own axis in the fork
- 145.** The nose of the aircraft is raised. How does the horizon ball of the ADI answer that movement?
- it is held still while the symbol climbs
  - it is swung the opposite way round by the fork
  - it spins about its own axis inside the fork
- 146.** A pilot flies the flight director bars while watching the horizon ball on the same indicator. What do the two indications give him together? ●
- where the aircraft is now and how fast it is turning
  - how the aircraft is banked and where the runway lies
  - where the aircraft is now and where it is to be flown
- 147.** On an ILS approach the runway symbol is seen rising towards the aeroplane symbol. What is it telling the crew?
- the radio altitude is decreasing
  - the localizer deviation is increasing
  - the glideslope is being captured
- 148.** An ADI is being used as the aircraft's rate of turn indicator. Which movement feeds that indication and what always accompanies it? ●
- the yaw rate, accompanied by the flight director bars
  - the roll rate, accompanied by the ball
  - the yaw rate, accompanied by the ball
- 149.** A technician presses the ATT TEST button and the ATT flag comes into view. Is the indicator faulty?
- no, null monitoring brings the flag in during the test
  - yes, the flag should stay out of view throughout the test
  - yes, the flag shows the test voltage never reached the synchros
- 150.** An ADI horizon ball has swung round to a 90° bank attitude. Which condition can produce this on some horizon indicators?
- the power supply has failed or the valid signal is missing
  - the attitude test button has been pressed and held in
  - the expanded part of the pitch scale has been run off

- 151.** An ADI's two synchro systems null correctly and its gyro reports valid, but the indicator has lost its power supply. Can the attitude still be displayed properly?
- yes, the synchros hold the ball in its last position
  - yes, the valid signal alone keeps the display alive
  - no, it is one of the conditions that must all be met
- 152.** An ADI loses the valid signal from its vertical gyro in flight. What can be expected of the attitude indication?
- the flag stays out of view and the ball holds its last attitude
  - the flag comes into view and the ball settles back level
  - the flag comes into view and the ball may tilt right over
- 153.** Two attitude indications disagree and the crew must work out which of them to believe. What does that always need? ●
- a repeat of the attitude comparison against the first source
  - the valid signal taken from each of the two gyros
  - a third independent source of the same information
- 154.** The two attitude indications disagree and vertical gyro number 3 is unserviceable. Which independent source is left for fault isolation? ●
- the weather radar system
  - the autoflight system
  - the standby horizon

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### Answers — Attitude director indicators

- 105. c** The page gives 115 V AC as the voltage supplied to the vertical gyro. (L1, PDF p. 681)
- 106. a** The page gives about 2 - 3 minutes for the fast erection phase. (L1, PDF p. 681)
- 107. b** The gyro is remote from the indicator and sits in the electric compartment. (L1, PDF p. 683)
- 108. b** Two independent gyros are needed, one for each pilot's indication. (L1, PDF p. 685)
- 109. a** The page names the autoflight system and the weather radar system as further users of the attitude data. (L1, PDF p. 685)
- 110. c** The two main values are the attitude, shown as an artificial horizon, and the flight director indications. (L1, PDF p. 687)
- 111. b** The page places the ADI in the Basic T. (L1, PDF p. 687)
- 112. b** The slip indicator is purely mechanical: a ball in a liquid filled curved glass tube. (L1, PDF p. 689)
- 113. a** The page gives 180°/min as the rate of turn movement represented. (L1, PDF p. 689)
- 114. b** It is there as an additional safety feature. (L1, PDF p. 693)
- 115. a** Fast erection is run only when the gyro has been without current for more than 30 seconds. (L2, PDF p. 681)
- 116. c** Those are exactly the conditions that end fast erection and drop the motor to reduced voltage. (L2, PDF p. 681)
- 117. b** Longitudinal acceleration cuts out the pitch erector, horizontal acceleration cuts out the roll erector. (L2, PDF p. 681)
- 118. a** The page limits the pitch erector cut-off to a maximum of three minutes. (L2, PDF p. 681)
- 119. c** The flag monitor requires VG power greater than 90 V before the valid voltage is given out. (L2, PDF p. 681)
- 120. c** Synchros on the gimbals convert the mechanical pitch and roll angles into electrical signals. (L2, PDF p. 683)
- 121. c** The valid signal is sent only while the gyro is operating at normal speed. (L2, PDF p. 683)
- 122. c** The valid signal is withheld until the gyro is up to normal speed and erection is complete, while the synchro signals are produced throughout. (L2, PDF p. 683)
- 123. b** A switching brings the third vertical gyro onto the indicator that lost its own gyro. (L2, PDF p. 685)
- 124. c** The third gyro can be switched to whichever of the two ADIs has lost its own source. (L2, PDF p. 685)
- 125. a** Being part of the housing, the symbol moves with the airframe it is bolted to. (L2, PDF p. 687)
- 126. c** The ball runs counter to the aircraft and the vertical gyro holds it level. (L2, PDF p. 687)

- 127. b The valid direct current is the gyro's statement that it is working. (L2, PDF p. 687)
- 128. c Gravity alone settles the ball, which is why it shows the apparent vertical. (L2, PDF p. 689)
- 129. b The yaw rate is the signal used for that indication. (L2, PDF p. 689)
- 130. b Its movement answers changes of rotational velocity about the vertical axis. (L2, PDF p. 689)
- 131. b It is a moving coil instrument. (L2, PDF p. 689)
- 132. a The localizer marks are about one degree of deviation each. (L2, PDF p. 689)
- 133. c The glideslope pointer is a moving coil instrument. (L2, PDF p. 689)
- 134. a The bars command the direction to fly, not the direction being flown. (L2, PDF p. 689)
- 135. b The signal is derived directly from the indication rather than from the source gyro. (L2, PDF p. 693)
- 136. c A difference of several degrees is what triggers the alert. (L2, PDF p. 693)
- 137. a The instrument warning system is the one that triggers the alert. (L2, PDF p. 693)
- 138. c The message is CHECK ATTITUDE and it appears on top of the attitude indication on both displays. (L2, PDF p. 693)
- 139. c The third reference is either vertical gyro number 3 or the standby horizon. (L2, PDF p. 693)
- 140. a Reporting valid marks the end of fast erection, and from then on the erectors run at the reduced rate of 2°/min. (L3, PDF p. 681)
- 141. c Invalid reporting together with the higher erector voltage is the signature of the fast erection phase. (L3, PDF p. 681)
- 142. c Of the four flag monitor conditions only completion of fast erection is still outstanding in this case. (L3, PDF p. 681)
- 143. a A duplicated system such as a second autopilot takes its attitude from the first officer's gyro, the first system taking the captain's. (L3, PDF p. 685)
- 144. a Roll is answered by the fork, which swings the ball the opposite way to the aircraft. (L3, PDF p. 687)
- 145. c Pitch is answered by the ball itself rotating about its own axis inside the fork. (L3, PDF p. 687)
- 146. c The attitude half states the present attitude and the flight director half commands the attitude to fly, so together they give present state and demand. (L3, PDF p. 689)
- 147. a The symbol doubles as the radio altimeter indication, and its travel up towards the aeroplane means the radio altitude is falling. (L3, PDF p. 689)
- 148. c The turn indication is driven by yaw rate and is always paired with the ball's slip indication. (L3, PDF p. 689)
- 149. a The flag is expected during the test because null monitoring brings it in. (L3, PDF p. 691)
- 150. a That deliberate tilt is the response to a lost power supply or a missing valid signal. (L3, PDF p. 691)
- 151. c All four listed conditions must hold together, and the power supply is one of them. (L3, PDF p. 691)
- 152. c Losing the valid signal removes the attitude display valid condition, so the flag appears, and on some indicators the ball is deliberately tilted right over. (L3, PDF p. 691)
- 153. c Two disagreeing sources can never settle themselves; a third independent source is always needed. (L3, PDF p. 693)
- 154. c The two named third sources are vertical gyro number 3 and the standby horizon, so with the gyro out the standby horizon remains. (L3, PDF p. 693)

## Turn and slip indicators

PDF pages 695–710 • 42 questions

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### LEVEL 1 — RECALL

155. Which turn must the aircraft be able to make in one minute when flying a holding pattern ●
- a. a 90° turn
  - b. a 360° turn
  - c. a 180° turn
156. Which sensor is used to measure how fast the aircraft is turning ●
- a. a rate gyro
  - b. an inclinometer
  - c. a vertical gyro
157. What damps the movement of the metal ball in the slip indicator ●
- a. a magnetic brake
  - b. a liquid
  - c. a light return spring
158. Of what material is the ball of the slip indicator made ●
- a. glass
  - b. metal
  - c. plastic
159. Which turn gives a full deflection of the rate of turn indication ●
- a. a half turn that would take one minute
  - b. a complete turn that would take two minutes
  - c. a complete turn that would take one minute
160. What is a complete turn flown in two minutes called ●
- a. a standard turn
  - b. a holding turn
  - c. a coordinated turn
161. Which degrees of freedom does the rate of turn indicator have ●
- a. one complete and one constrained
  - b. two complete and none constrained
  - c. one complete and two constrained
162. To which aircraft axis is the gyroscope axis of the rate of turn indicator parallel ●
- a. the vertical axis
  - b. the longitudinal axis
  - c. the lateral axis
163. How many degrees of freedom does a rate gyro have ●
- a. three
  - b. two
  - c. one

**LEVEL 2 — UNDERSTAND AND EXPLAIN**

- 164.** Which three indicators can be used to fly the required rate of turn ●
- the turn and slip indicator, the turn coordinator and the ADI
  - the turn and slip indicator, the turn coordinator and the standby horizon
  - the turn and slip indicator, the vertical gyro and the ADI
- 165.** An aircraft is fitted with a glass cockpit. How is the rate of turn presented to its crew? ●
- on a pointer over a mechanical scale
  - on a display
  - by the ball of the inclinometer
- 166.** Which instrument gives the slip indication that always goes with the rate of turn indication ●
- an inclinometer, also called the ball
  - a rate gyro, also called the turn needle
  - a vertical gyro, also called the horizon
- 167.** What does the slip indication show the pilot during a turn
- the direction of the bank
  - the direction of the resultant forces
  - the direction of the rate of turn
- 168.** What do the rate of turn pointer and the ball tell the pilot when they are read together
- the radius of the turn
  - the bank angle of the turn
  - the quality of the turn
- 169.** The ball of a slip indicator sits exactly in the centre during a turn. Why does that mean the turn is perfectly coordinated? ●
- the vertical axis is parallel to the resultant force
  - the vertical axis is parallel to the true vertical
  - the rate of turn has reached its full deflection
- 170.** In which direction are the rudder pedals moved whenever the ball has run out of centre ●
- towards the direction of the turn
  - away from the ball
  - towards the ball
- 171.** The gyro of a turn indicator precedes against its return spring. On what does that precession force depend? ●
- on the gyro stability, which follows the gyro mass
  - on the stiffness of the return spring itself
  - on the gyro stability, which follows the gyro speed
- 172.** About which aircraft axis can the frame carrying the gyro axis rotate
- about the vertical axis
  - about the lateral axis
  - about the longitudinal axis
- 173.** By how much is the rate gyro of a turn coordinator turned from its normal position ●
- by 90 degrees
  - by 20 degrees
  - by 45 degrees

- 174.** Why is the rate gyro of a turn coordinator canted ●
- a. so that it is also sensitive to the pitch rate
  - b. so that it is also sensitive to the roll rate
  - c. so that it is no longer sensitive to the yaw rate
- 175.** For what is the rate gyro mainly used ●
- a. as a stable long term attitude reference
  - b. as a highly sensitive rotary motion sensor
  - c. as a damper for the slip indicator ball
- 176.** A force acts on the constrained input axis of a rate gyro. How does the gyro answer it? ●
- a. it precesses around the input axis
  - b. it tilts bodily along the spin axis
  - c. it precesses around the precession axis
- 177.** By which means can the second degree of freedom of a rate gyro be limited ●
- a. by a spring or by a mechanical stop
  - b. by a counterweight or by magnetic braking
  - c. by a spring or by movement dependent damping
- 178.** Which input to output behaviours can a rate gyro be built to have ●
- a. integral, differential or exponential
  - b. proportional, integral or differential
  - c. proportional, linear or logarithmic
- 179.** What is often used in place of the spring in a rate of turn gyro ●
- a. a torsion bar
  - b. a damping piston
  - c. a torque motor
- 180.** About how many aircraft axes can a rate of turn gyro measure rotary velocity ●
- a. about two, set by its two degrees of freedom
  - b. about all three, whatever its orientation
  - c. about one only, set by the way it is fitted
- 181.** How are angular velocity gyroscopes built with integrated features constrained ●
- a. by viscose damping rather than a spring
  - b. by a torsion bar rather than a spring
  - c. by a spring rather than by damping
- 182.** Which gyro measures angles in the aircraft axes with very high precision ●
- a. an angular velocity gyroscope with integrated features
  - b. an attitude gyro with three degrees of freedom
  - c. a directional gyroscope with automatic drift compensation
- 183.** Towards what are devices inside the aircraft drawn when the aircraft slips ●
- a. towards the outside of the curve
  - b. towards the nose of the aircraft
  - c. towards the centre of the curve

**LEVEL 3 — APPLY AND REASON**

- 184.** In a turn the ball has moved out towards the inside of the turn. What does that say about the aircraft? ●
- the bank angle is too small for the actual rate of turn
  - the bank angle suits the rate of turn but the speed is too high
  - the bank angle is too large for the actual rate of turn
- 185.** The ball has moved out in the direction of the turn. What must the pilot do to bring the turn back into balance? ●
- raise the rate of turn with rudder towards the ball
  - lower the bank angle and leave the rudder central
  - lower the rate of turn with rudder towards the ball
- 186.** The ball has moved out opposite to the direction of the turn. What is wrong and what has to be done? ●
- the bank angle is too small, so the rate of turn must be raised
  - the bank angle is too large, so the rate of turn must be raised
  - the bank angle is too small, so the rate of turn must be lowered
- 187.** The needle shows exactly the turn the crew want but the ball lies well out of centre. What do the two indications say about the turn together? ●
- the rate is wrong but the turn is coordinated
  - the rate is right but the turn is not coordinated
  - the rate and the coordination are both right
- 188.** The gyro of a turn and slip indicator is running below its normal speed. What happens to the rate of turn indication? ●
- it loses accuracy because the precession force falls away
  - it stays accurate because the spring alone sets the deflection
  - it loses accuracy because the spring force falls away
- 189.** The turn needle of a turn and slip indicator stands at full deflection while the ball is centred. What is the aircraft doing? ●
- turning at a rate that would complete a circle in one minute, in balance
  - turning at a rate that would complete a circle in two minutes, out of balance
  - turning at a rate that would complete a circle in two minutes, in balance
- 190.** The aircraft starts to turn about its vertical axis. What then happens inside the rate of turn indicator?
- the frame is held still and the gyro axis tilts the pointer directly
  - the gyro axis stays put while the frame is turned by the return spring
  - the gyro axis precedes and twists the frame against the return spring
- 191.** A rate of turn indicator is built with only one free and one constrained degree of freedom and a return spring. What can it therefore show? ●
- the attitude the aircraft is holding
  - the bank angle the aircraft has reached
  - how fast the aircraft is turning

- 192.** An aircraft is holding a steady bank angle through a turn. What is the turn coordinator showing at that moment? ●
- the yaw rate
  - the bank angle itself
  - the roll rate
- 193.** A pilot is rolling into a turn and the bank angle is still changing. What is the turn coordinator showing while that goes on? ●
- the roll rate
  - the yaw rate
  - the resultant force of the turn
- 194.** Two aircraft are rolled into a bank at the same moment, one carrying a rate of turn indicator and one a turn coordinator. Whose indication answers the roll in? ●
- the one whose gyro lies along the lateral axis
  - the one whose gyro lies along the vertical axis
  - the one whose gyro is canted by 45 degrees
- 195.** A rate of turn gyro is held by a linear spring. What can then be said about its precession angle? ●
- it is proportional to the angular velocity about the measuring axis
  - it is independent of the angular velocity about the measuring axis
  - it falls as the angular velocity about the measuring axis rises
- 196.** During a turn the loose items in the cabin are being pushed out towards the outside of the curve. Which condition is that? ●
- a slip
  - a skid
  - a perfectly coordinated turn

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### Answers — Turn and slip indicators

- 155. c** The holding pattern turn is defined as 180° in one minute. (L1, PDF p. 695)
- 156. a** The rate of turn sensor is a rate gyro. (L1, PDF p. 695)
- 157. b** The ball's movement is damped by the liquid it sits in. (L1, PDF p. 697)
- 158. b** The ball is a metal one. (L1, PDF p. 697)
- 159. b** Full deflection corresponds to a complete turn in two minutes. (L1, PDF p. 699)
- 160. a** The manual names that the standard turn. (L1, PDF p. 699)
- 161. a** It has one complete degree of freedom and one that is constrained. (L1, PDF p. 701)
- 162. c** The gyro axis lies parallel to the lateral axis. (L1, PDF p. 701)
- 163. b** A rate gyro has only two degrees of freedom. (L1, PDF p. 705)
- 164. a** The three named are the turn and slip indicator, the turn coordinator and the ADI. (L2, PDF p. 695)
- 165. b** In a glass cockpit the rate of turn appears on a display. (L2, PDF p. 695)
- 166. a** The slip indication comes from an inclinometer, known as the ball. (L2, PDF p. 695)
- 167. b** It shows which way the resultant forces lie during the turn. (L2, PDF p. 695)
- 168. c** Read together they state how good the turn is. (L2, PDF p. 697)
- 169. a** A centred ball means the aircraft's vertical axis is parallel to the resultant force of the turn. (L2, PDF p. 697)
- 170. c** In both of the out of balance cases the pedals go in the direction of the ball. (L2, PDF p. 697)
- 171. c** The precession force depends on gyro stability, and that stability is proportional to the speed. (L2, PDF p. 699)
- 172. c** The frame's limited rotation is about the longitudinal axis. (L2, PDF p. 701)

- 173. c** The gyro is canted by 45 degrees. (L2, PDF p. 703)
- 174. b** Canting the gyro adds sensitivity to roll rate. (L2, PDF p. 703)
- 175. b** Its main use is as a very sensitive sensor of rotary motion. (L2, PDF p. 705)
- 176. c** A force on the input axis is answered by precession about the precession axis. (L2, PDF p. 705)
- 177. c** The page gives a spring or movement dependent damping as the ways of limiting it. (L2, PDF p. 705)
- 178. b** The three behaviours named are proportional, integral and differential. (L2, PDF p. 705)
- 179. a** A torsion bar often replaces the spring. (L2, PDF p. 705)
- 180. b** All three aircraft axes can be measured, and the gyro's orientation does not matter. (L2, PDF p. 705)
- 181. a** They use viscose damping in place of a spring. (L2, PDF p. 705)
- 182. a** The page states that these gyros measure angles of attitude in the aircraft axes with very high precision, in contrast to attitude gyros. (L2, PDF p. 705)
- 183. c** In a slip the devices inside are drawn in towards the centre of the curve. (L2, PDF p. 709)
- 184. c** A ball out in the direction of the turn means too much bank for the rate of turn being flown. (L3, PDF p. 697)
- 185. a** With the ball out into the turn the turn rate must be raised, using rudder towards the ball. (L3, PDF p. 697)
- 186. c** A ball out the other way means too little bank for the rate flown, and the cure is to lower the rate of turn. (L3, PDF p. 697)
- 187. b** The needle sets the rate and the ball sets the coordination, so a good needle with a displaced ball is a turn at the right rate flown out of balance. (L3, PDF p. 697)
- 188. a** Low gyro speed means low stability and so a weakened precession force, which is why the indication is accurate only at normal speed. (L3, PDF p. 699)
- 189. c** Full deflection is the two minute turn and a centred ball is a coordinated one, so both are being flown. (L3, PDF p. 699)
- 190. c** The turn about the vertical axis makes the gyro precede, and that precession twists the frame against the spring. (L3, PDF p. 701)
- 191. c** With the second freedom held by a spring the gyro can only balance precession against spring force, which measures how fast the aircraft turns, not the attitude it holds. (L3, PDF p. 701)
- 192. a** With the bank angle constant the coordinator reads yaw rate. (L3, PDF p. 703)
- 193. a** While the bank angle is changing the coordinator reads roll rate instead. (L3, PDF p. 703)
- 194. c** Only the turn coordinator's gyro is canted, and that is what makes it answer roll rate while the bank angle is changing. (L3, PDF p. 703)
- 195. a** A linear spring makes the precession angle proportional to the angular velocity about the measuring axis. (L3, PDF p. 705)
- 196. b** With the loose items thrown outward from the centre of the curve the ball also lies to the outside, which Figure 208 labels a skid. (L3, PDF p. 710)

## Directional gyros

PDF pages 711–719 · 30 questions

### LEVEL 1 — RECALL

- 197.** What other name is given to the direction indicator ●
- Position indicator
  - Heading indicator
  - Attitude indicator
- 198.** How many degrees of freedom and which axis does a direction indicator have ●
- two degrees of freedom and a horizontal axis
  - three degrees of freedom and a horizontal axis
  - three degrees of freedom and a vertical axis
- 199.** Which three effects does the indication precision of a gyro compass depend on ●
- Apparent drift, inclination and cardan error
  - Apparent drift, real drift and cardan error
  - Apparent drift, real drift and parallax error
- 200.** A directional gyro has just been synchronised. For how long does its indication stay relatively exact? ●
- 15 - 20 minutes
  - 5 - 10 minutes
  - 5 - 10 seconds
- 201.** What are directional gyros used as in remote compass systems ●
- Indicators
  - Sensors
  - Amplifiers
- 202.** Which two parts carry out the erection task in an electrical directional gyro ●
- A pressure switch and a torque motor
  - A level switch and a flux valve
  - A level switch and a torque motor
- 203.** Which task is the liquid level switch with its torque motor fitted for ●
- Drift elimination
  - Levelling
  - Monitoring
- 204.** A directional gyro component list includes a slaving torque motor. What is that motor fitted for? ●
- To eliminate drift
  - To monitor the rpm
  - To level the frame
- 205.** Which two kinds of drift does the slaving torque motor eliminate ●
- Real drift and pole drift
  - Apparent drift and real drift
  - Apparent drift and cardan error

**LEVEL 2 — UNDERSTAND AND EXPLAIN**

- 206.** Why does the indication stay steady while the aircraft turns its nose ●
- The gyro keeps its own direction stable
  - The gyro is driven faster during the turn
  - The gyro is locked to the airframe during the turn
- 207.** A direction indicator has just reached its operational speed. What has to happen before its heading can be used? ●
- The heading must be set automatically
  - The heading must be set from the lubber line
  - The heading must be set manually
- 208.** What property of a space stable gyro allows it to hold the course
- It maintains the position it has taken up
  - It aligns itself with the earth's magnetic lines
  - It returns to the horizontal after every turn
- 209.** What must the rotational axis of a gyro compass remain parallel to for its directional information to be read ●
- The earth's surface
  - The aircraft's longitudinal axis
  - The earth's magnetic axis
- 210.** How is a gyro compass set and from which indication ●
- Manually, from the flux valve
  - Manually, from the magnetic compass
  - Automatically, from the magnetic compass
- 211.** Where are the sensor and the indicator of a remote compass system installed ●
- The sensor in the cockpit and the indicator anywhere
  - The sensor anywhere and the indicator in the cockpit
  - Both the sensor and the indicator in the cockpit
- 212.** An erection system is fitted to a directional gyro. In which plane does it hold the gyro axis?
- Horizontal
  - Vertical
  - Along the lubber line
- 213.** What is the important difference between a directional gyro and a vertical gyro ●
- The directional gyro is never electrical
  - The directional gyro has no erection system
  - The directional gyro drifts
- 214.** A commercial aircraft is fitted with electrical directional gyros. How is their drift dealt with? ●
- It is compensated automatically
  - It is compensated by the vertical gyro
  - It is compensated by the pilot each hour
- 215.** Why is the directional gyro more sophisticated than the gyro compass ●
- It is part of the remote compass system, not of the panel
  - It is part of the standby compass, not of the panel
  - It is part of the instrument panel, not of a remote system

- 216.** The inner frame of a directional gyro leaves the horizontal. What does the liquid level switch signal? ●
- The heading synchro
  - The torque motor
  - The slaving amplifier
- 217.** To which axis does the levelling torque motor give momentum ●
- The axis of the inner frame
  - The axis of the gyro rotor
  - The axis of the outer frame
- 218.** What produces the signals received by the slaving torque motor ●
- The earth magnetic field detector
  - The heading synchro amplifier
  - The liquid level switch circuit
- 219.** A directional gyro is running up after power-on. What rotor speed must be reached before a valid signal is passed?
- 50%
  - 75%
  - 100%

### LEVEL 3 — APPLY AND REASON

- 220.** A pilot is about to adjust the direction indicator after start-up. Which instrument has to be read first? ●
- The attitude indicator
  - The turn indicator
  - The magnetic compass
- 221.** A crew finds the gyro compass reading has moved away from the compass after some minutes of flight. What does the drift of the instrument call for? ●
- Re-caging the outer frame by hand at regular intervals
  - Synchronising the course indication by hand at intervals
  - Re-levelling the inner frame by hand at regular intervals
- 222.** A new course has to be taken up during a turn lasting a couple of minutes. Why does the gyro compass suit this task? ●
- Its indication is corrected from magnetic north throughout
  - Its indication stays exact over such a short period
  - Its indication is free of drift while the aircraft turns
- 223.** A heading sensor is mounted well away from the instrument panel and its reading appears on the flight deck. Which gyro is the sensor and what fault must be corrected in it? ●
- A directional gyro, whose toppling must be compensated
  - A directional gyro, whose drift must be compensated
  - A vertical gyro, whose drift must be compensated
- 224.** A directional gyro in a remote compass system is found to have no slaving torque motor fitted. What does this mean for its drift?
- It cannot be compensated at all
  - It is removed by the levelling torque motor
  - It is compensated by other means

- 225.** A directional gyro has begun to drift and the slaving momentum is applied to the inner frame axis. Until what point does it continue? ●
- Until the gyro heading matches the standby compass heading
  - Until the gyro heading matches the flux valve heading
  - Until the inner frame is back in the horizontal
- 226.** The servo system of the indicator does not compare correctly. What does the monitoring system of the directional gyro do?
- It passes no valid signal to the consumer
  - It passes a valid signal to the consumer anyway
  - It passes the flux valve heading instead

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### Answers — Directional gyros

- 197. b** The page gives heading indicator as the second name for the direction indicator. (L1, PDF p. 711)
- 198. b** The direction indicator has three degrees of freedom together with a horizontal axis. (L1, PDF p. 711)
- 199. b** The page lists exactly these three as the limits on precision. (L1, PDF p. 713)
- 200. b** The printed figure for the useful period is 5 - 10 minutes. (L1, PDF p. 713)
- 201. b** In a remote compass system the directional gyro is the sensing element. (L1, PDF p. 715)
- 202. c** The electrical gyro erects with a level switch driving a torque motor. (L1, PDF p. 715)
- 203. b** The level switch and its torque motor are the levelling pair. (L1, PDF p. 717)
- 204. a** The slaving torque motor is listed against drift elimination. (L1, PDF p. 717)
- 205. b** The slaving torque motor is fitted against apparent and real drift. (L1, PDF p. 719)
- 206. a** The gyro holds its direction, so the scale moves relative to the aircraft and shows the new heading. (L2, PDF p. 711)
- 207. c** Once the gyro is up to speed the heading has to be adjusted by hand. (L2, PDF p. 711)
- 208. a** Holding the position it was set to is what keeps the course indication. (L2, PDF p. 713)
- 209. a** The page ties the readable indication to the axis staying parallel to the earth's surface. (L2, PDF p. 713)
- 210. b** The gyro compass is set by hand to match the magnetic compass. (L2, PDF p. 713)
- 211. b** Remote means the sensing unit can sit anywhere while only the display is in the cockpit. (L2, PDF p. 715)
- 212. a** For a directional gyro the erection system holds the axis horizontal. (L2, PDF p. 715)
- 213. c** Drift is the behaviour that sets the directional gyro apart from the vertical gyro. (L2, PDF p. 715)
- 214. a** Electrical gyros in commercial aircraft carry automatic drift compensation. (L2, PDF p. 715)
- 215. a** Being a remote system component rather than a panel instrument is the reason given for its greater complexity. (L2, PDF p. 717)
- 216. b** The level switch output goes to the torque motor. (L2, PDF p. 719)
- 217. c** Levelling momentum is applied to the outer frame axis until the inner frame is horizontal. (L2, PDF p. 719)
- 218. a** The slaving signals come from an earth magnetic field detector. (L2, PDF p. 719)
- 219. b** The monitoring condition printed for rotor speed is 75%. (L2, PDF p. 719)
- 220. c** The setting value comes from the magnetic compass, which the direction indicator cannot supply itself. (L3, PDF p. 711)
- 221. b** Because apparent drift, real drift and cardan error spoil the precision, the course indication has to be resynchronised by hand against the magnetic compass. (L3, PDF p. 713)
- 222. b** The short-term accuracy of the gyro is enough for the few minutes a turn onto a new heading takes. (L3, PDF p. 713)
- 223. b** The remote sensor is the directional gyro, and the page states its drift must be compensated. (L3, PDF p. 715)
- 224. c** The page states drift can also be compensated by other means in remote compass systems. (L3, PDF p. 717)
- 225. b** Slaving runs until the gyro agrees with the flux valve, which is the magnetic reference of the system. (L3, PDF p. 719)
- 226. a** A correct servo comparison is one of the conditions for a valid signal, so failing it blocks the signal. (L3, PDF p. 719)

# Terrestrial magnetism and the direct reading compass

PDF pages 723–739 · 44 questions

## LEVEL 1 — RECALL

- 227.** In which direction do the compass systems always point ●
- a. Magnetic north
  - b. Compass north
  - c. True north
- 228.** What is the angle between true north and magnetic north called ●
- a. Inclination
  - b. Deviation
  - c. Variation
- 229.** What represents the aircraft longitudinal axis in a heading indication ●
- a. The compass scale
  - b. The lubber line
  - c. The magnetic needle
- 230.** Which magnetic field lines can a compass use to measure the heading ●
- a. The declining ones
  - b. The vertical ones
  - c. The horizontal ones
- 231.** Where are all the magnetic field lines horizontal ●
- a. Near the isogonic lines
  - b. Near the equator
  - c. Near the poles
- 232.** Which three methods are used to measure the angle between the aircraft longitudinal axis and magnetic north ●
- a. Mechanical, electronic and mathematical
  - b. Mechanical, electrical and magnetic
  - c. Electronic, mathematical and gyroscopic
- 233.** Which electrical part does a direct reading compass contain ●
- a. A torque motor
  - b. A detector coil
  - c. A light
- 234.** Where is the standby compass usually located ●
- a. At the centerpost between the front windshields
  - b. On the centre panel below the front windshields
  - c. On the overhead panel above the front windshields
- 235.** On which law of magnetism is the function of the magnetic compass based ●
- a. All poles attract when they are close enough
  - b. Unlike poles attract and like poles repel
  - c. Like poles attract and unlike poles repel

- 236.** What kind of fluid is used in the compass housing to inhibit the movement of the card ●
- A mineral or alcohol based fluid
  - A water or glycol based fluid
  - A silicone or ester based fluid
- 237.** To what does the aircraft magnetic field change the direction of magnetic north ●
- Grid north
  - True north
  - Compass north
- 238.** What is the difference between magnetic north and compass north called ●
- Deviation
  - Variation
  - Inclination

## LEVEL 2 — UNDERSTAND AND EXPLAIN

- 239.** A magnet has magnetic lines running between its poles. What is the effective area of those lines called? ●
- The magnetic field
  - The magnetic pole
  - The magnetic axis
- 240.** How may the earth be regarded in terms of its magnetism ●
- As a giant electromagnet with a wound coil interior
  - As a giant permanent magnet with a bar magnet interior
  - As a giant permanent magnet with a hollow interior
- 241.** What does a suspended, free-turning magnetic needle align itself with ●
- The earth's magnetic axis
  - The earth's magnetic lines
  - The aircraft's longitudinal axis
- 242.** An aircraft is flown to a different part of the world. What does the actual value of the variation depend on?
- The position on earth
  - The aircraft's own magnetic field
  - The heading being flown
- 243.** Between which two directions is the indicated heading measured ●
- The lubber line and the aircraft longitudinal axis
  - The aircraft longitudinal axis and the direction to the north pole
  - The aircraft lateral axis and the direction to the north pole
- 244.** Why are the different variations at different locations basically no problem
- Navigation equipment removes the local variation automatically
  - Navigation equipment, runways and maps all use the magnetic heading
  - Navigation equipment, runways and maps all use the true heading
- 245.** Why must runway headings be updated every few years ●
- The variation is zero along the isogonic lines
  - The position of the magnetic poles is drifting
  - The position of the geographic poles is drifting

246. The earth's magnetic field is being resolved at a given latitude. Into which two components is it split?
- a. A horizontal and a lateral component
  - b. A horizontal and a vertical component
  - c. A magnetic and a geographic component
247. An aircraft measures its heading by the electronic method. What determines the angle? ●
- a. A freely rotating permanent magnet
  - b. An earth magnetic field detector
  - c. A stored variation value
248. Which compass name is given to the method that uses an earth magnetic field detector ●
- a. Remote reading compass
  - b. Standby reading compass
  - c. Direct reading compass
249. Why is the direct reading compass called the standby compass in commercial aircraft ●
- a. Its indication replaces the gyro indication during every turn
  - b. Its indication belongs to the basic T and must always be available
  - c. Its indication belongs to the basic T and is the primary one in flight
250. Why is the standby compass placed at the centerpost ●
- a. Both pilots can read it and foreign magnetic fields are distant
  - b. Both pilots can reach it and direct sunlight is avoided
  - c. Only one pilot can read it but foreign magnetic fields are distant
251. The scale of a standby compass takes up the direction of the field. What turns it? ●
- a. A permanent magnet
  - b. An electromagnet
  - c. A torque motor
252. Why is the direct reading compass filled with a fluid ●
- a. To insulate the compass scale from heat
  - b. To dampen the compass scale movements
  - c. To lubricate the compass scale bearings
253. With what does the freely moving magnet of a compass line up ●
- a. The horizontal components of the earth's field
  - b. The longitudinal axis of the aircraft
  - c. The vertical components of the earth's field
254. What does the compass card under the lubber line show ●
- a. The magnetic field against the aircraft's longitudinal axis
  - b. The longitudinal axis against the earth's geographic pole
  - c. The longitudinal axis against the earth's magnetic pole
255. What condition must be met for the lubber line to represent the aircraft longitudinal axis ●
- a. The compass must be correctly filled
  - b. The compass must be correctly compensated
  - c. The compass must be correctly installed

256. The earth's magnetic field is weak. What does this mean for the compass magnet?
- A high directive force of the compass magnet
  - A low directive force of the compass magnet
  - A high inclination of the compass magnet
257. The damping medium changes volume with temperature. What balances this out? ●
- Slots
  - Washers
  - Bellows
258. What inside the aircraft also affects the direct reading compass ●
- A vibration field from electrical equipment
  - A magnetic field from the earth's interior
  - A magnetic field from electrical equipment
259. A standby compass is found to have a compass error from the airframe. What is used to minimize it? ●
- Small adjustable weights inside the compass
  - Small adjustable magnets inside the compass
  - Copper washers under the mounting plate

### LEVEL 3 — APPLY AND REASON

260. Why does the north pole of the magnetic needle show north ●
- The geographic south pole of the earth is there
  - The magnetic south pole of the earth is there
  - The magnetic north pole of the earth is there
261. An aircraft carries a modern inertial reference unit. Why is a compass reference carried as well? ●
- Inertial units are accurate for a short time only
  - Inertial units cannot produce a heading at all
  - Radio indications are referenced to magnetic north
262. A chart gives the variation for a position and the compass card carries a deviation. Which pair of directions does each of these angles lie between? ●
- Variation between true and compass north, deviation between true and magnetic north
  - Variation between true and magnetic north, deviation between magnetic and compass north
  - Variation between magnetic and compass north, deviation between true and magnetic north
263. An airport publishes a changed runway designation a few years after the previous one. What has made this necessary? ●
- The magnetic poles have moved and magnetic references must follow
  - The compass deviation at the airfield has changed
  - The geographic poles have moved and true references must follow
264. An aircraft is to operate in a region close to the magnetic pole. Which heading source must be used there and why? ●
- A flux valve, because the vertical component takes over
  - A directional gyro, because the horizontal component is too weak
  - A standby compass, because the horizontal component is strongest

- 265.** What does growing inclination mean for the indication of a magnetic compass ●
- The vertical component weakens and the indication becomes unusable
  - The horizontal component weakens and the indication becomes unusable
  - The horizontal component grows and the indication becomes unstable
- 266.** A compass finds the angle with a mounted, freely rotating permanent magnet. Which method and which compass is this? ●
- The mechanical method, a direct reading compass
  - The electronic method, a remote reading compass
  - The mechanical method, a remote reading compass
- 267.** An aircraft finds its heading from data held in a computer rather than by sensing the field. Which method is this and what does it use? ●
- The electronic method, using the field detector fitted at each location
  - The mathematical method, using the stored variation for each location
  - The mechanical method, using the magnet mounted at each location
- 268.** An aircraft suffers a total electrical power loss. Which heading indication is still available? ●
- The horizontal situation indicator
  - The directional gyro
  - The standby compass
- 269.** Fluid damping has been accepted so the indicator stays stable under vibration. What has to be accepted along with it?
- An increase of the installation error
  - A loss of directive force of the magnet
  - A time delay after a change of direction
- 270.** New electrical equipment has been fitted close to the standby compass. Which compass error will change and what corrects it? ●
- Deviation, taken out by the compensation magnets
  - Variation, taken out by the compensation magnets
  - Deviation, taken out by updating the chart value

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### Answers — Terrestrial magnetism and the direct reading compass

- 227. a** Compass systems follow the earth's magnetic field, which defines magnetic north. (L1, PDF p. 723)
- 228. c** Variation is the printed name for the true-to-magnetic north angle. (L1, PDF p. 725)
- 229. b** The lubber line stands for the longitudinal axis of the aircraft. (L1, PDF p. 725)
- 230. c** Only the horizontal field lines are usable for heading measurement. (L1, PDF p. 731)
- 231. b** The field lines run horizontal only in the equatorial region. (L1, PDF p. 731)
- 232. a** The three headings printed on the page are mechanical, electronic and mathematical. (L1, PDF p. 733)
- 233. c** The only electrical item in a direct reading compass is its light. (L1, PDF p. 735)
- 234. a** The usual position is the centerpost between the two front windshields. (L1, PDF p. 735)
- 235. b** The main law of magnetism quoted is that unlike poles attract and like poles repel. (L1, PDF p. 737)
- 236. a** The damping medium named is a mineral or alcohol based fluid. (L1, PDF p. 737)
- 237. c** The shifted direction is named compass north. (L1, PDF p. 739)
- 238. a** Deviation is the printed name for that difference. (L1, PDF p. 739)
- 239. a** The area over which the lines act is the magnetic field. (L2, PDF p. 723)
- 240. b** The page describes the earth as a giant permanent magnet whose interior is a bar magnet. (L2, PDF p. 723)

- 241. b** A free needle takes up the direction of the earth's magnetic lines, which is what makes it usable for navigation. (L2, PDF p. 723)
- 242. a** Variation is a property of where the aircraft is, not of how it is flown. (L2, PDF p. 725)
- 243. b** Heading is the angle from the aircraft longitudinal axis to north. (L2, PDF p. 725)
- 244. b** Everything is referenced the same way, to magnetic heading, so the local value does not have to be applied. (L2, PDF p. 727)
- 245. b** Because the magnetic poles move, anything referenced to them goes out of date. (L2, PDF p. 727)
- 246. b** The field is resolved into horizontal and vertical components. (L2, PDF p. 731)
- 247. b** The electronic method senses the field with an earth magnetic field detector. (L2, PDF p. 733)
- 248. a** A detector-based system is named the remote reading compass. (L2, PDF p. 733)
- 249. b** Being part of the basic T, the heading has to remain available at all times, which is the standby role. (L2, PDF p. 735)
- 250. a** The position suits both crew members and keeps the compass clear of equipment fields. (L2, PDF p. 735)
- 251. a** The scale is driven by a permanent magnet aligning with the field. (L2, PDF p. 735)
- 252. b** The fluid is there as a damping medium for the scale. (L2, PDF p. 735)
- 253. a** The compass magnet follows the horizontal components of the earth's field. (L2, PDF p. 737)
- 254. c** The reading at the lubber line is the longitudinal axis direction against the earth's magnetic pole. (L2, PDF p. 737)
- 255. c** The lubber line only stands for the longitudinal axis if the installation is correct. (L2, PDF p. 737)
- 256. b** A weak field gives the compass magnet only a weak directive force. (L2, PDF p. 737)
- 257. c** Bellows take up the expansion and contraction of the damping fluid. (L2, PDF p. 737)
- 258. c** Equipment running in normal operation sets up a magnetic field inside the aircraft. (L2, PDF p. 739)
- 259. b** The compass compensation consists of small adjustable magnets inside the instrument. (L2, PDF p. 739)
- 260. b** What lies at the earth's north is a magnetic south pole, and unlike poles attract. (L3, PDF p. 723)
- 261. c** Even with INS or IRS fitted, indications such as the VOR RMI are referenced to magnetic north. (L3, PDF p. 723)
- 262. b** Variation separates true from magnetic north and deviation separates magnetic north from compass north. (L3, PDF p. 725)
- 263. a** Runway directions are magnetic references, so pole drift forces them to be re-published. (L3, PDF p. 727)
- 264. b** Inclination leaves too little horizontal field at high latitude, so a gyro or inertial reference replaces the compass. (L3, PDF p. 731)
- 265. b** Inclination takes the field lines out of the horizontal, and the horizontal component the compass needs is halved by 50 degrees of latitude and unusable beyond 70 degrees. (L3, PDF p. 731)
- 266. a** The magnetic needle principle is the mechanical method and gives the direct reading compass. (L3, PDF p. 733)
- 267. b** Working from stored variation values in a navigation computer is the mathematical method. (L3, PDF p. 733)
- 268. c** The direct reading standby compass needs no electrical power to show the heading. (L3, PDF p. 735)
- 269. c** The page calls the delay an unavoidable disadvantage of keeping the indicator steady. (L3, PDF p. 737)
- 270. a** An added aircraft field alters the deviation, and deviation is reduced by the compass compensation magnets. (L3, PDF p. 739)

## Compass compensation and swinging

PDF pages 741–755 • 27 questions

### LEVEL 1 — RECALL

- 271.** A compass is about to be compensated — into which classes are the errors to be removed divided? ●
- A, B and C errors
  - B, C and D errors
  - A, B and D errors
- 272.** A compass is found to carry a B error — in which directions does it show up? ●
- in the north-south direction
  - in the east-west direction
  - in all directions
- 273.** An engineer is picking out tools for a compass compensation — which ones may be used? ●
- only approved antimagnetic tools
  - only insulated electrical tools
  - only calibrated torque tools
- 274.** Within what tolerance must the aircraft be aligned to North - South before the SCC is used? ●
- exactly  $\pm 5$  degrees
  - exactly  $\pm 10$  degrees
  - exactly  $\pm 2$  degrees
- 275.** What is required of the deviation that is left once a compensation has been performed? ●
- it should be as small as possible
  - it should be exactly zero on every heading
  - it should be below one degree on every heading
- 276.** In what steps are the headings checked when a deviation card is made up? ●
- in steps of 45 degrees
  - in steps of 30 degrees
  - in steps of 10 degrees
- 277.** An aircraft is flying over the northern part of the earth — in which direction does the vertical component of the field point there? ●
- downwards
  - upwards
  - along the meridian
- 278.** On which flight direction does an acceleration or a deceleration leave the heading indication unchanged? ●
- a westerly direction
  - a northerly direction
  - an easterly direction

### LEVEL 2 — UNDERSTAND AND EXPLAIN

- 279.** Why does a magnetic compass or a remote compass system have to be compensated at all?
- aircraft ferro- and electro-magnetism sets up fields that cause a deviation
  - the vertical field component pulls the magnet out of the horizontal plane
  - the earth's field loses strength as the aircraft climbs away from the ground

- 280.** Some error is still left after a compass has been compensated — where is that remainder written down? ●
- on the compensating disc at the airport site
  - on a deviation card carried on the aircraft
  - on a label fitted beside the flux valve
- 281.** An indication is falsified by the amount of the deviation — what is that heading called? ●
- the true heading
  - the magnetic heading
  - the compass heading
- 282.** An engineer walks out to compensate a compass — which items that he is carrying can deflect it?
- his notebook, his pen and his pass
  - his ear defenders, his gloves and his torch
  - his glasses, his watch and his mobile phone
- 283.** What does the Standby Compass Calibrator allow to be done without physically turning the aircraft ●
- compensate permanent magnetisation errors and complete the deviation card
  - measure the vertical component of the earth's field at the compass
  - set the directional gyro to agree with the remote compass heading
- 284.** Which two separate functions does the SCC perform
- it cancels the local earth's field and adds a rotatable standard field
  - it rotates the standby compass and records the resulting deviation
  - it cancels the aircraft's permanent field and adds a vertical field
- 285.** The aircraft cannot be put exactly on North - South for a standby compass calibration — what is done about the residual misalignment? ●
- it is entered on the deviation card
  - it is taken out on the E-W screw
  - it is measured and set into the SCC
- 286.** A compensation is finished and a small deviation is still there — why can it not be driven to zero?
- the card is only read in steps of thirty degrees
  - the earth's field changes while the aircraft is turned
  - that is nearly impossible with the means for adjustment
- 287.** The readings have been taken on every checked heading — what is done with each of them? ●
- each one is remarked on a deviation card
  - each one is remarked in the compensation log
  - each one is set into the remote compensator
- 288.** A compass magnet is kept level on its bearing — which component of the earth's field does it then measure?
- the vertical component
  - the total resultant field
  - the horizontal component
- 289.** What decides whether a tilted compass turns to the right or to the left ●
- the actual bank angle of the aircraft
  - the actual airspeed of the aircraft
  - the actual heading of the aircraft

290. A compass has been tilted and the vertical component is acting on its magnet — what does the crew see? ●
- a frozen heading
  - a falsified heading
  - a slowly spinning card
291. An aircraft decelerates while flying on a southerly heading — what happens to the heading indication? ●
- it swings towards the east
  - it swings towards the west
  - it is not influenced
292. Why is there no heading error when the aircraft accelerates on a northerly heading? ●
- the vertical field has no component at all at that position on the earth
  - the vertical field acts on the magnet the way the horizontal field does
  - the magnet is held exactly horizontal by its bearing on that heading
293. Why does a southerly heading behave like a northerly one during an acceleration? ●
- the pivot point bearing locks the compass scale
  - the permanent magnet points in the same direction
  - the vertical component reverses with the hemisphere

### LEVEL 3 — APPLY AND REASON

294. Compensation is being worked from INS/IRS data alongside the aircraft's own heading sources — which source supplies the magnetic heading?
- the INS/IRS
  - the standby compass
  - the remote compass system
295. A remote compass is 4 ° off the target course on one of the four main headings — what does that mean for the job?
- it is in tolerance and the 30° headings can be started
  - it is in tolerance for a main heading but not for a 30° heading
  - it is out of tolerance and must be adjusted further
296. No compensating site is free and the aircraft cannot be turned — which method still works? ●
- a bearing plate sighted on a distant object
  - the markings on the compensating disc
  - the Standby Compass Calibrator
297. An acceleration has tilted a compass out of the horizontal plane — what now acts on its magnet? ●
- the vertical component of the earth's magnetic field
  - the horizontal component of the earth's magnetic field
  - the permanent magnetism of the aircraft structure

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### Answers — Compass compensation and swinging

271. a The manual classes the compensated errors as A, B and C. (L1, PDF p. 741)
272. b A B error is the east-west error. (L1, PDF p. 741)
273. a Compensation may only be done with approved antimagnetic tools. (L1, PDF p. 743)
274. c The printed alignment requirement is exactly +/- 2 degrees of North - South. (L1, PDF p. 749)

- 275. a** The requirement is that the deviation is as small as possible. (L1, PDF p. 751)
- 276. b** Headings are checked every 30 degrees. (L1, PDF p. 751)
- 277. a** Over the northern part the vertical component points down. (L1, PDF p. 753)
- 278. b** On a northerly heading an acceleration or deceleration leaves the indication alone. (L1, PDF p. 755)
- 279. a** Interference fields from the aircraft's own ferro- and electro-magnetism produce the deviation. (L2, PDF p. 741)
- 280. b** The remaining deviation goes on the deviation card that stays on the aircraft. (L2, PDF p. 741)
- 281. c** The falsified indication is named the compass heading. (L2, PDF p. 741)
- 282. c** Glasses, a watch and a mobile phone are the items named as causing deflection. (L2, PDF p. 743)
- 283. a** The SCC compensates permanent magnetisation errors and completes the deviation card with the aircraft standing still. (L2, PDF p. 747)
- 284. a** The SCC cancels the local earth's field and adds a standard field that can be rotated. (L2, PDF p. 749)
- 285. c** The misalignment is measured and fed into the SCC itself. (L2, PDF p. 749)
- 286. c** Zero cannot practically be reached with the available means of adjustment. (L2, PDF p. 751)
- 287. a** Each checked heading's reading goes onto the deviation card. (L2, PDF p. 751)
- 288. c** Holding the magnet horizontal lets it measure the horizontal component. (L2, PDF p. 753)
- 289. c** The direction of the swing depends on the heading the aircraft is on. (L2, PDF p. 753)
- 290. b** The result at the instrument is a falsified heading. (L2, PDF p. 753)
- 291. c** South behaves like north, so the indication is not influenced. (L2, PDF p. 755)
- 292. b** Both field components pull the magnet the same way, so nothing moves. (L2, PDF p. 755)
- 293. b** On south the permanent magnet still points the same way, so the same reasoning holds. (L2, PDF p. 755)
- 294. c** The remote compass gives magnetic heading, the INS/IRS gives true heading. (L3, PDF p. 743)
- 295. c** The remote compass tolerance is  $+/- 2^\circ$ , so  $4^\circ$  is outside it and more adjustment is needed. (L3, PDF p. 745)
- 296. c** The SCC works without rotating the aircraft, while the other two methods both need the aircraft turned on a site. (L3, PDF p. 747)
- 297. a** Once the compass is off the horizontal the vertical component starts to influence the magnet. (L3, PDF p. 753)

# The flux valve and remote compass systems

PDF pages 757–767 • 15 questions

## LEVEL 1 — RECALL

- 298.** A remote compass system is being explained — which two sensing devices does it use? ●
- a. a sensor for the aircraft's own field and a rate gyro
  - b. a sensor for the earth's magnetic field and a directional gyro
  - c. a sensor for the earth's magnetic field and a vertical gyro
- 299.** Where is the indicator of a remote compass system fitted? ●
- a. in a wing tip fairing
  - b. in the avionics compartment
  - c. in the instrument panel
- 300.** What does the flux valve turn the direction of the magnetic field into? ●
- a. a proportional electrical signal
  - b. a proportional change of resistance
  - c. a proportional mechanical rotation

## LEVEL 2 — UNDERSTAND AND EXPLAIN

- 301.** A large aeroplane carries two separate remote compass systems — how are they shared out?
- a. one for each pilot
  - b. one for each gyro platform
  - c. one for each engine
- 302.** What does the signal detected by the flux valve represent? ●
- a. the angle between the directional gyro and the aircraft's longitudinal axis
  - b. the strength of the earth's vertical field at the aircraft's longitudinal axis
  - c. the direction of the earth's horizontal field to the aircraft's longitudinal axis
- 303.** A flux valve has to stay level as the aircraft manoeuvres — how is it suspended in its housing? ●
- a. on three rubber mounts
  - b. on a needle point bearing
  - c. fully cardanically
- 304.** On what do the intensity and the direction of the sensing coil signals depend
- a. on how fast the aircraft is turning
  - b. on how strong the excitation current is
  - c. on where the magnetic field points
- 305.** Under which condition does a coil induce a voltage at all? ●
- a. only when the intensity of the magnetic field changes
  - b. only when the intensity of the magnetic field is highest
  - c. only when the iron core has reached full saturation
- 306.** How is the iron core of the flux valve made to change its magnetic state? ●
- a. it is frequently turned by a small motor inside the housing
  - b. it is frequently demagnetised by a direct current in a sensing coil
  - c. it is frequently saturated by an alternating current in an excitation coil

- 307.** What gives the flux valve its name ●
- the liquid that flows through its hemisphere
  - the vent valve that breathes with altitude
  - the alternate switching of the magnetic field

### LEVEL 3 — APPLY AND REASON

- 308.** An aircraft is held in a 40° banked turn — what happens to the flux valve measuring plane? ●
- it is driven back to horizontal by the directional gyro
  - it stays horizontal because the cardan takes any angle
  - it can no longer stay horizontal
- 309.** The excitation current has just fallen away and the core has come out of saturation — what does the magnetic field do?
- it collapses completely until the core saturates again
  - it takes the shortest way and bypasses the three coils
  - it takes an even longer way through the iron and the coils
- 310.** The excitation coil of a flux valve has gone open circuit — what reaches the magnetic heading reference system? ●
- a signal that gives true heading instead of magnetic
  - nothing, because the field intensity no longer changes
  - a signal at half of its normal output frequency
- 311.** How does the sinusoidal voltage pattern of a flux valve differ from that of an ordinary synchro?
- it is shifted near to the zero value of the excitation voltage
  - it is clipped at the saturation level of the iron core
  - it is shifted near to the peak value of the excitation voltage
- 312.** A compass coupler is getting no setting signal at all — which chain has broken down?
- the flux valve to control transformer path
  - the compass coupler to excitation coil feed path
  - the directional gyro to flux valve drive path

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### Answers — The flux valve and remote compass systems

- 298. b** The system pairs an earth's field sensor with a directional gyro. (L1, PDF p. 757)
- 299. c** The indicator sits in the instrument panel. (L1, PDF p. 757)
- 300. a** The output is an electrical signal proportional to the field direction. (L1, PDF p. 759)
- 301. a** The two systems are split one per pilot. (L2, PDF p. 757)
- 302. c** The flux valve signal gives the direction of the earth's horizontal field relative to the aircraft's longitudinal axis. (L2, PDF p. 757)
- 303. c** It hangs in a full cardan suspension. (L2, PDF p. 757)
- 304. c** Both the intensity and the direction follow the direction of the magnetic field. (L2, PDF p. 759)
- 305. a** A changing field intensity is what induces the voltage. (L2, PDF p. 759)
- 306. c** An alternating current in the excitation coil repeatedly saturates the core. (L2, PDF p. 759)
- 307. c** The field is switched on and off like a valve, which is where the name comes from. (L2, PDF p. 759)
- 308. c** The cardan only holds the plane level out to about 30°, so at 40° of bank it is exceeded. (L3, PDF p. 757)
- 309. c** Unsaturated iron offers a far easier path, so the field runs the longer way through the iron and the coils. (L3, PDF p. 759)

- 310. b** Without the excitation the core is never switched in and out of saturation, so the field intensity at the coils is steady and no voltage is induced. (L3, PDF p. 759)
- 311. a** In the flux valve the pattern sits near the zero value of the excitation voltage. (L3, PDF p. 767)
- 312. a** The printed path runs flux valve to control transformer to the directional gyro or compass coupler, so a loss at the coupler points back along that path. (L3, PDF p. 767)